



Lab02: Languages and Operation on Languages

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Group Odd number: For Mirror of Languages

1. Why does $(L_1 \cup L_2)^R = L_1^R \cup L_2^R$ hold? Can you show an example to illustrate it?
2. How does the mirror of a concatenation $(L_1.L_2)^R$ become $=L_2^R L_1^R$?
Why is the order reversed?
3. Why does taking the mirror twice $(L^R)^R$ return the original language L ?
4. What is the difference between $(L^+)^R$ and $(L^R)^+$?
5. What about $(L^*)^R$ and $(L^R)^*$? Why are they equal?

Group Even number: For Closure (Star and Plus)

1. Why is $L^0 = \{\varepsilon\}$ in the definition of language powers?
2. Why does the Kleene star L^* include ε but the positive closure L^+ does not?
3. Can you explain why $L^* = L^+ \cup L^0$?
4. Is it always true that $L^+ = L^* \setminus \{\varepsilon\}$? Provide proof or a counterexample.
5. Is the concatenation of two languages always commutative?
That is, is $L_1.L_2 = L_2.L_1$? Provide an example to support your answer.